COMPLETE SELF-SIMILAR SOLUTION OF THE SUBSONIC MARSHAK WAVE

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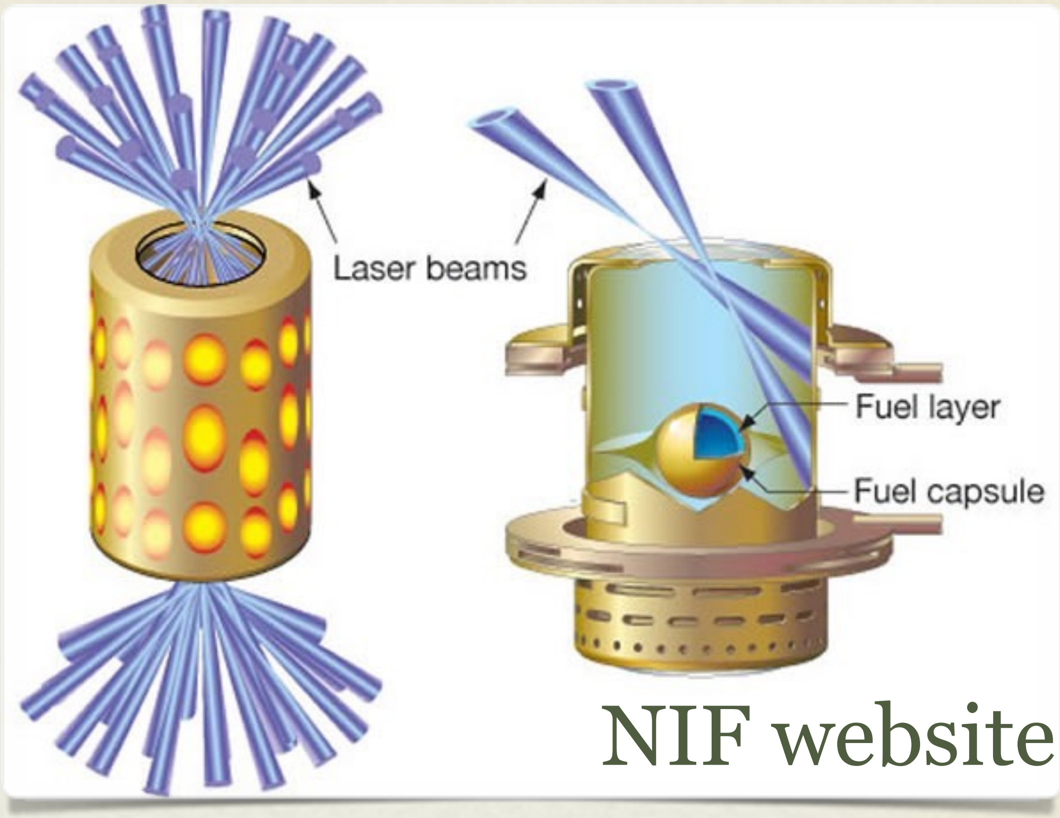
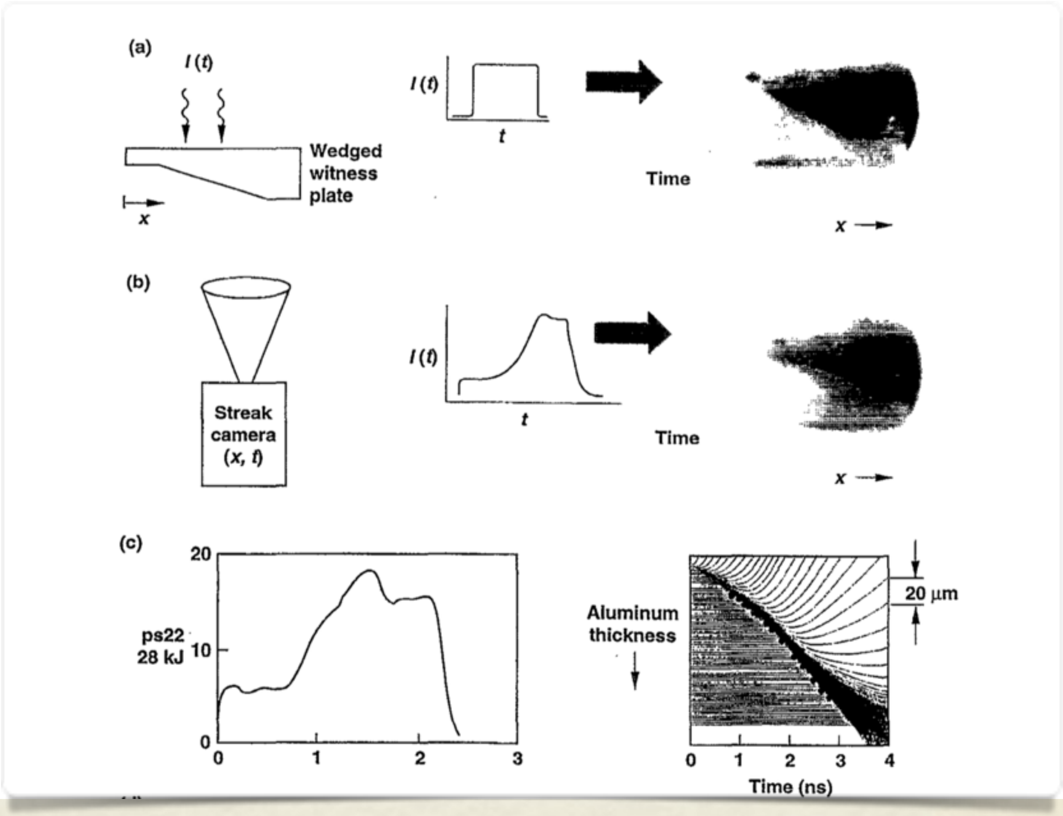
MOTIVATION

- ICF:
  - Hohlraum temperature prediction & measurement
  - Laser - Walls coupling analysis
  - Pulse shaping
  - [Lindl 1995], [Kauffman 1995], Rosen 2005]
- EOS Experiments at a varying pressure profile [Thomas 2006]



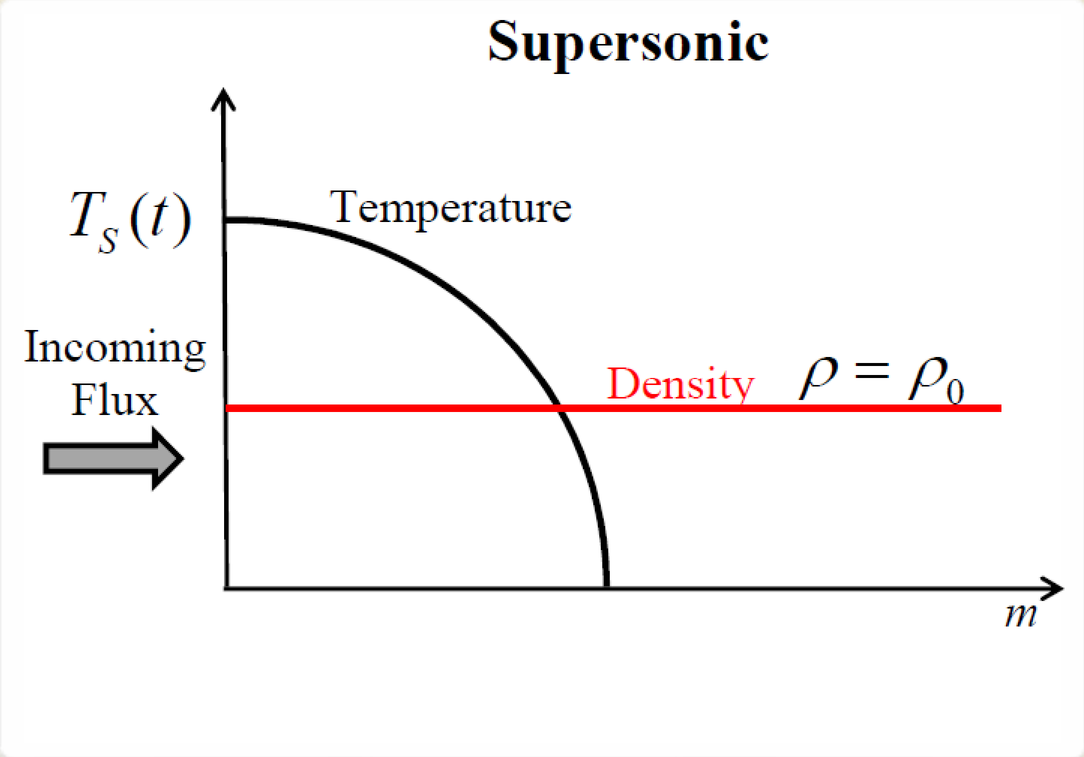
NIF website

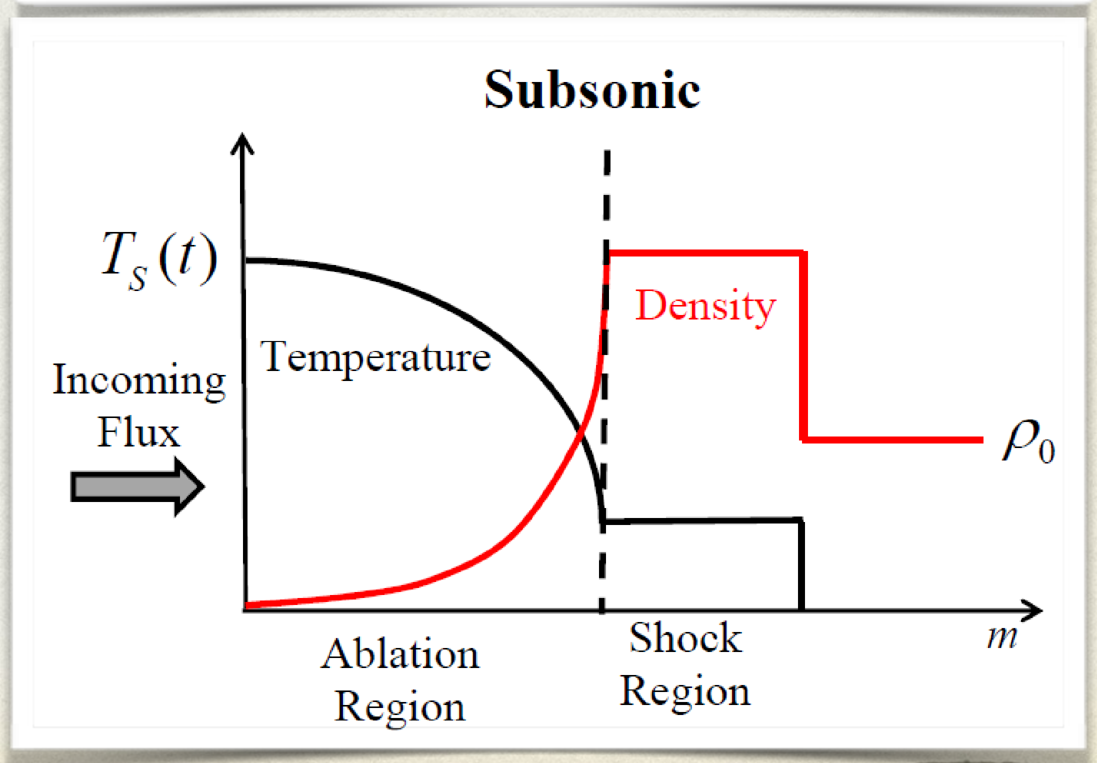


Lindl 1995

THE MECHANISM

- We consider a semi-infinite wall, heated to high temperature at the boundary.
- The wall heats via photon transport.
- In the optically thick limit, a sharp temperature front arises.
- If the heat front is slower than the speed of sound, a shock region propagates ahead of the ablation region.





PREVIOUS WORKS

- [R. E. Marshak 1958] - First description and super-sonic solutions.
- [R. Pakula & R. Sigel 1985] - Self-Similar solutions of the ablation region.
- [J.H Hammer & M.D. Rosen 2003] - New solutions based on a perturbation theory
- [J. Garnier et. al. 2006] - Solved the heat and the shock, for a specific boundary condition.
- None of the works above provided a full solution for both regions.

OUR WORK - OVERVIEW

- The idea: Solve each region separately and find a way to patch the solutions.
- For each region, we neglect one of the dimensional parameters, to make the problem self-similar.
- The heat front will serve as the boundary condition for the shock.

ASSUMPTIONS

- Boundary Condition: $$T(t) = T_0 t^{\tau}$$
- Opacity: $$1/\kappa_R = g T^{\alpha} \rho^{-\lambda}$$
- Heat Capacity: $$e = f T^{\beta} \rho^{-\mu}$$
- Equation of State: $$P = r\rho e \equiv (\gamma - 1)\rho e$$
- Initial Density: $$\rho_0$$

STATEMENT OF THE PROBLEM

- The hydro-radiation equations:

$$\frac{\partial V}{\partial t} = \frac{\partial u}{\partial m}$$

$$\frac{\partial u}{\partial t} = -\frac{\partial P}{\partial m}$$

$$\frac{\partial e}{\partial t} + P\frac{\partial V}{\partial t} = \frac{\partial}{\partial m}\left(\frac{c}{3\kappa_R}\frac{\partial (aT^4)}{\partial m}\right)$$

- The energy equation becomes:

$$\frac{1}{r}\frac{\partial PV}{\partial t} + P\frac{\partial V}{\partial t} = B\frac{\partial}{\partial m}\left(V^{\lambda}\frac{\partial}{\partial m}(PV^{1-\mu})^{\frac{4+\alpha}{\beta}}\right)$$

$$B = \frac{16\sigma}{3(4+\alpha)}g(rf)^{\frac{-4+\alpha}{\beta}}$$

THE ABLATION REGION

- The problem is characterised by four dimensional parameters:

$$B, T_0, \rho_0 (V_0), t$$

- For most the ablation region, the density is much lower than ρ_0.

- We assume the density diverges at the front.

THE ABLATION REGION

$$X = \tilde{X} B^{w_{X1}} T_0^{w_{X2}} t^{w_{X3}}$$

Physical variable	Power-law dependency
Lagrangian Coordinate	$m = \xi \left(B^{2-\mu} T_0^{8+2\alpha-2\beta+\beta\lambda-(4+\alpha)\mu} t^{2+2(4+\alpha-\beta\tau-\mu(3+(4+\alpha)\tau)+\lambda(2+\beta\tau)} \right)^{\frac{1}{4+2\lambda-4\mu}}$
Temperature	$T(m,t) = T_0 t^{\tau} \widetilde{T}$
Velocity	$u(m,t) = \widetilde{u} \left(B^{-\mu} T_0^{\beta(2+\lambda)-(4+\alpha)\mu} t^{\mu+\beta(2+\lambda)\tau-(4+\alpha)\mu\tau} \right)^{\frac{1}{4+2\lambda-4\mu}}$
Pressure	$P(m,t) = \widetilde{P} \left(B^{1-\mu} T_0^{4+\alpha+\beta\lambda-(4+\alpha)\mu} t^{-1+\mu+(4+\alpha+\beta\lambda)\tau-(4+\alpha)\mu\tau} \right)^{\frac{1}{2+\lambda-2\mu}}$
Specific Volume	$V(m,t) = \widetilde{V} \left(B^{-1} T_0^{-4-\alpha+2\beta} t^{1-(4+\alpha-2\beta)\tau} \right)^{\frac{1}{2+\lambda-2\mu}}$
Heat Flux	$S(m,t) = \widetilde{S} \left(B^{2-3\mu} T_0^{8+2\alpha+2\beta+3\lambda\beta-3(4+\alpha)\mu} t^{-2+3\mu+(2(4+\alpha+\beta)+3\beta\lambda)\tau-3(4+\alpha)\mu\tau} \right)^{\frac{1}{4+2\lambda-4\mu}}$
Energy	$E(m,t) = \widetilde{E} \left(B^{2-3\mu} T_0^{8+2\alpha+2\beta+3\lambda\beta-3(4+\alpha)\mu} t^{2+2\lambda-\mu+(2(4+\alpha+\beta)+3\beta\lambda)\tau-3(4+\alpha)\mu\tau} \right)^{\frac{1}{4+2\lambda-4\mu}}$

THE SELF SIMILAR EQUATION

- We obtain a dimensionless system of ordinary differential equations.
- The boundary conditions are:

$$\tilde{V}(\xi_F)=0, \frac{\partial \tilde{V}}{\partial \xi}|_{\xi_F}=0$$

$$\tilde{P}(0)=0, \frac{\partial \tilde{P}}{\partial \xi}\xrightarrow[\xi\to 0]{}\infty$$

$$\tilde{u}(\xi_F)=0$$

- For each value of ξ_F a unique solution exists.
- The true value is determined from the condition

$$\tilde{T}(0)=1$$

RESULTS

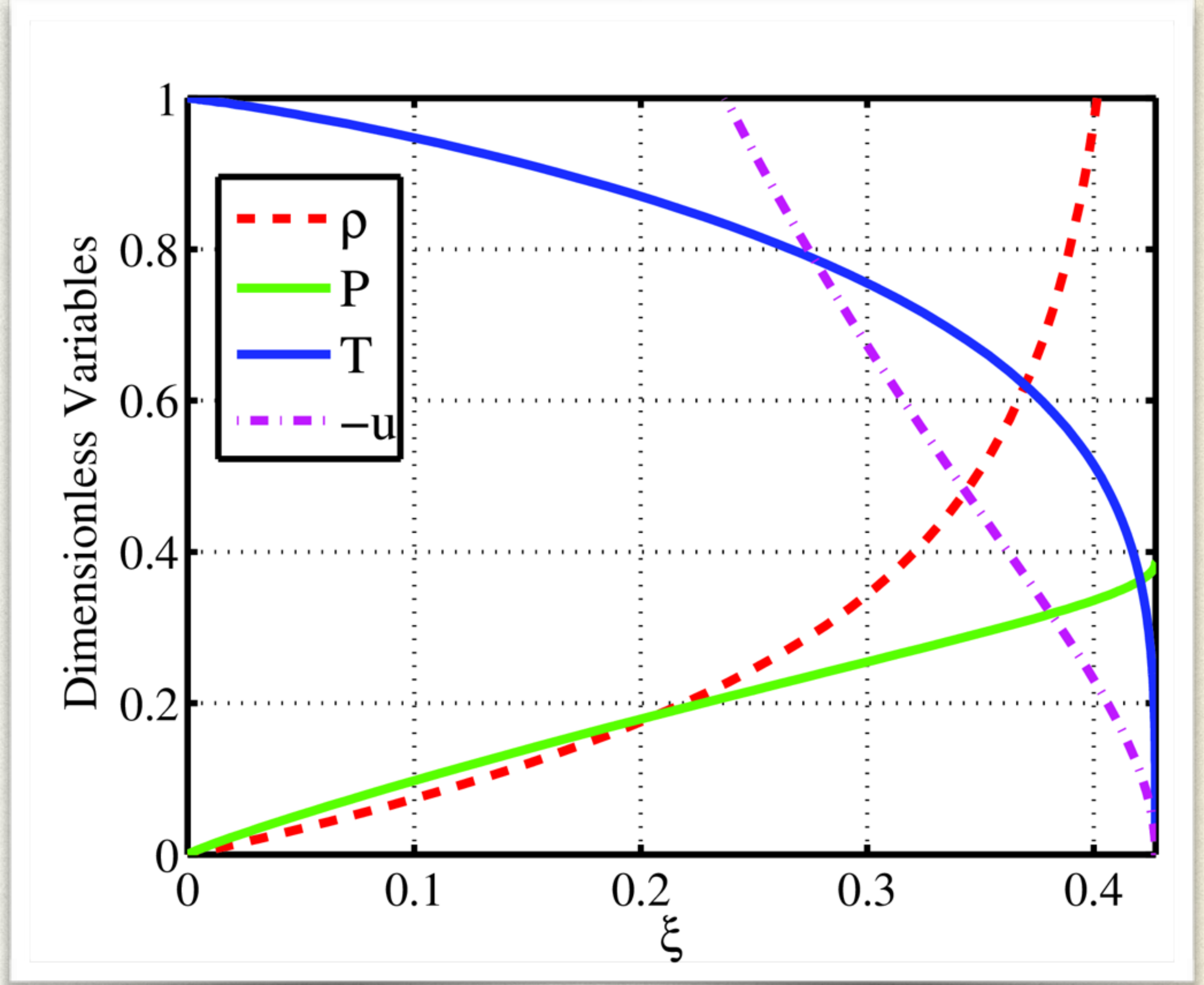
- For constant T:

$$m_F = 10.17 \cdot 10^{-4} T_0^{1.91} t^{0.52} [\frac{g}{cm^2}]$$

$$E = 0.59 T_0^{3.35} t^{0.59} [\frac{hJ}{mm^2}]$$

$$P_F = 2.71 T_0^{2.63} t^{-0.45} Mbar$$

- Similar relations can be obtained for different boundary conditions:
  - Constant Net-Flux
  - Constant Pressure
- P will be used as the shock boundary condition.



THE SHOCK REGION

- This time, we use the boundary condition: $$P(t) = P_0 t^{\tau_S}$$
- With strong shock boundary conditions at the front:

$$\frac{D-u}{V_s} = \frac{D}{V_0}$$

$$P_S = \frac{Du_S}{V_0}$$

- We neglect B (no flux at the shock region).

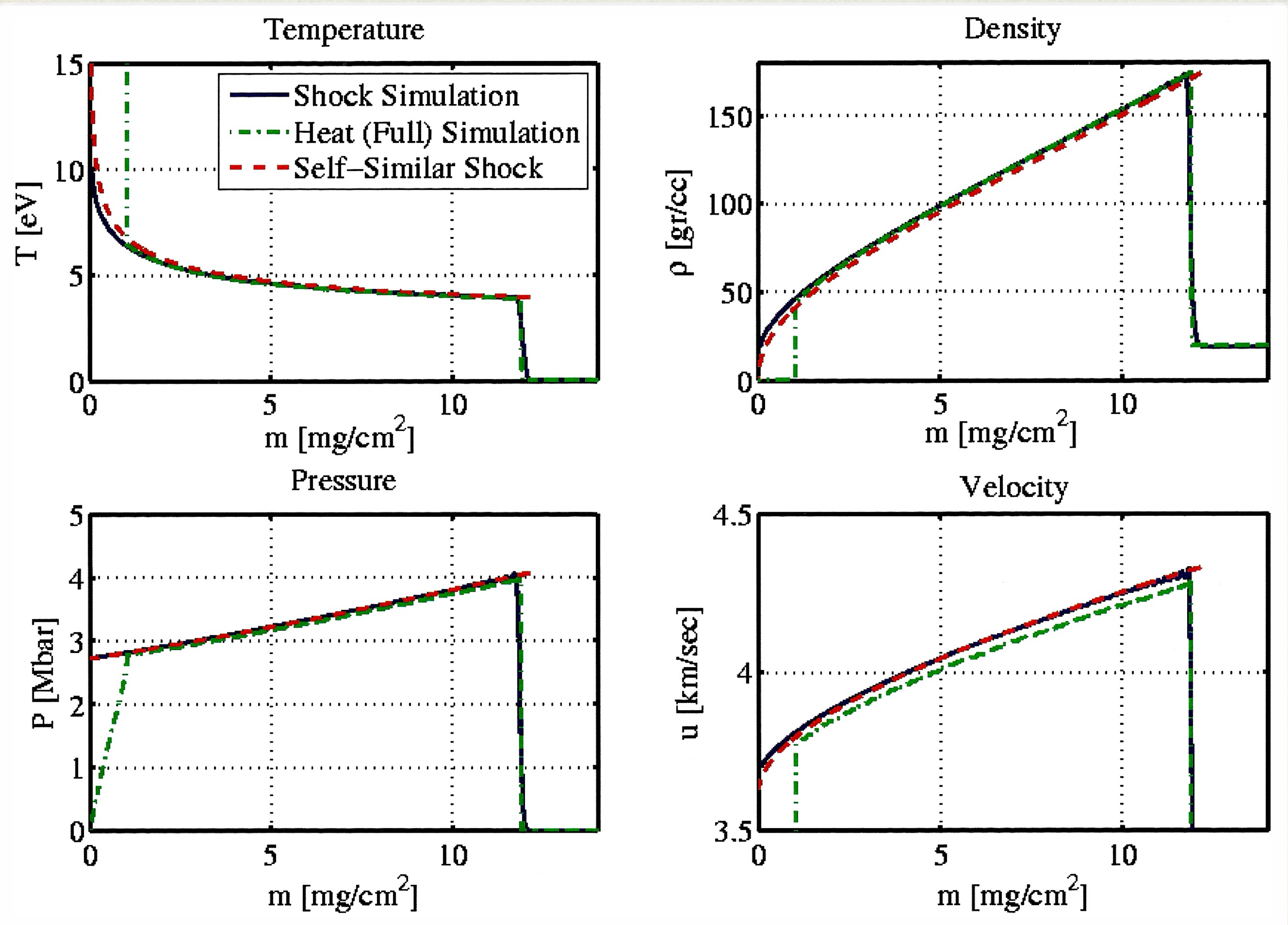
$$P_s u_s V_0 = D(e_S + \frac{u_s^2}{2})$$

- We note that the EOS could (and should) be different:

$$r \neq r'$$

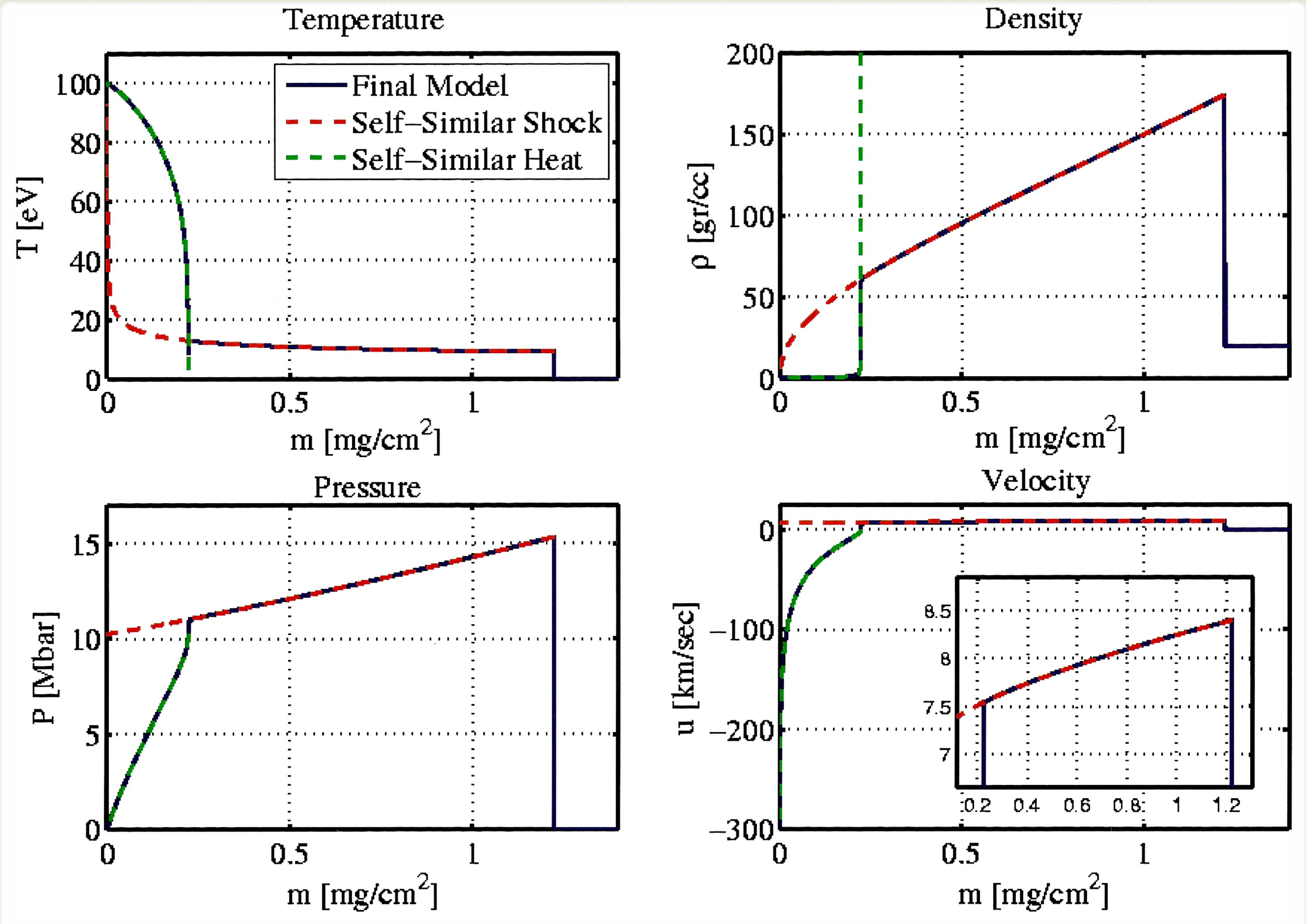
THE SELF SIMILAR EQUATION

- For the boundary condition $T_0 = 100eV, \tau = 0$
- And the corresponding shock condition $P_0 = 2.71Mbar$, $\tau_S = -0.45$

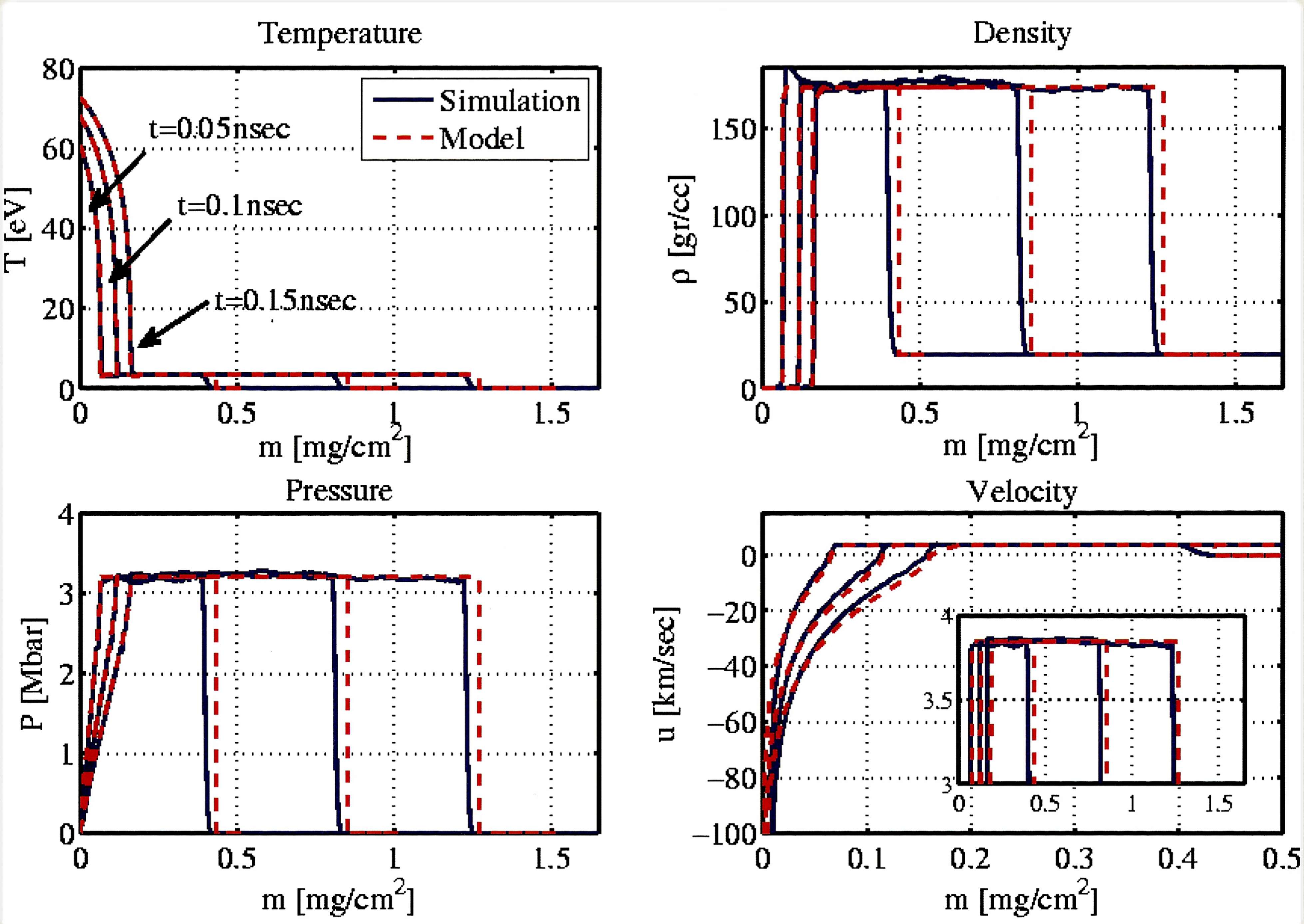


Physical variable	Power-law dependency
Lagrangian Coordinate	$m_S = \xi P_0^{\frac{1}{2}} V_0^{-\frac{1}{2}} t^{1+\frac{\tau_S}{2}}$
Velocity	$u(m,t) = \widetilde{u} P_0^{\frac{1}{2}} V_0^{\frac{1}{2}} t^{\frac{\tau_S}{2}}$
Pressure	$P(m,t) = \widetilde{P} P_0 t^{\tau_S}$
Specific Volume	$V(m,t) = \widetilde{V} V_0$
Energy	$E(m,t) = \widetilde{E} P_0^{\frac{3}{2}} V_0^{\frac{1}{2}} t^{1+\frac{3}{2}\tau_S}$

FULL SOLUTION



COMPARISON TO NUMERICAL SIMULATIONS



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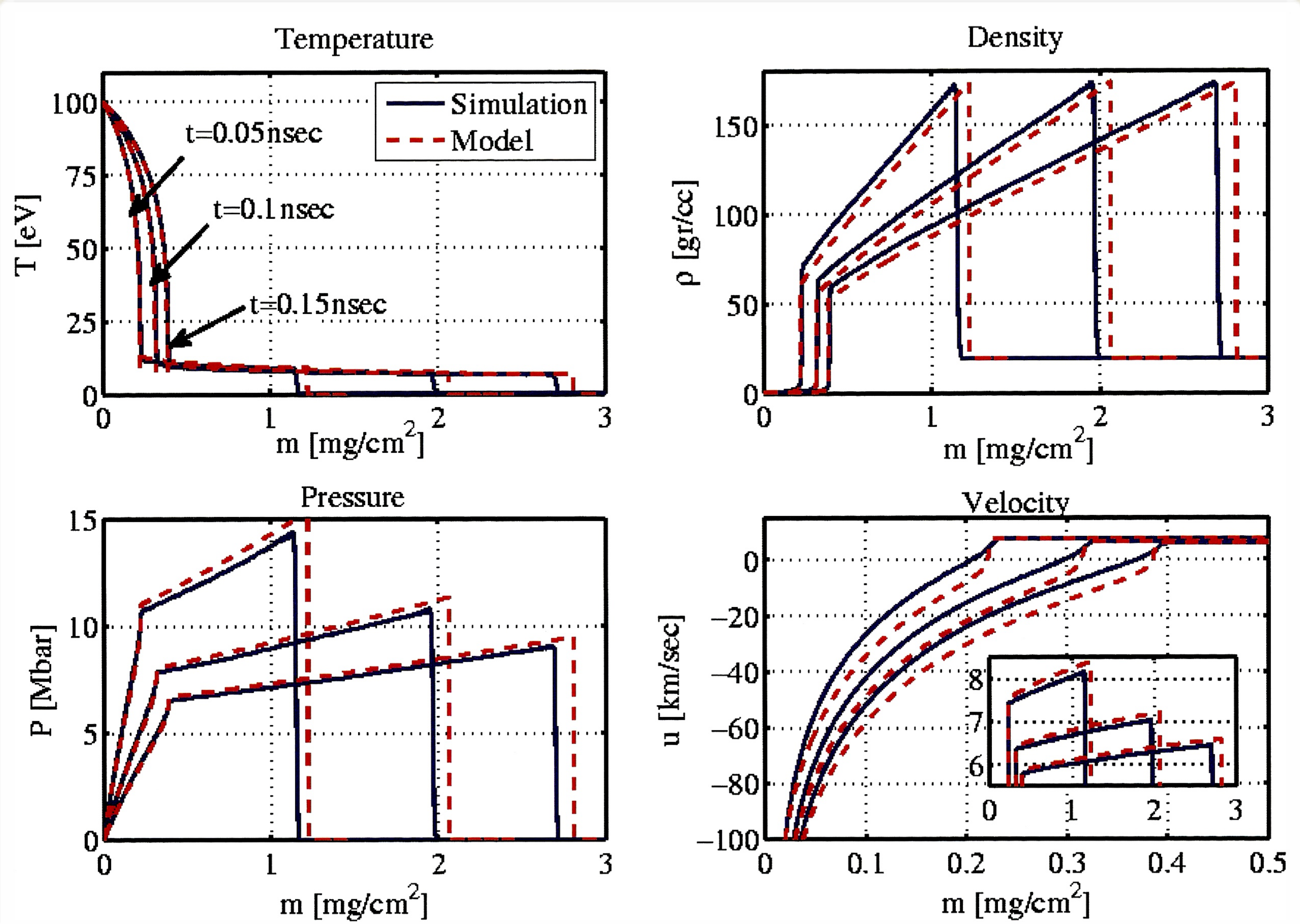

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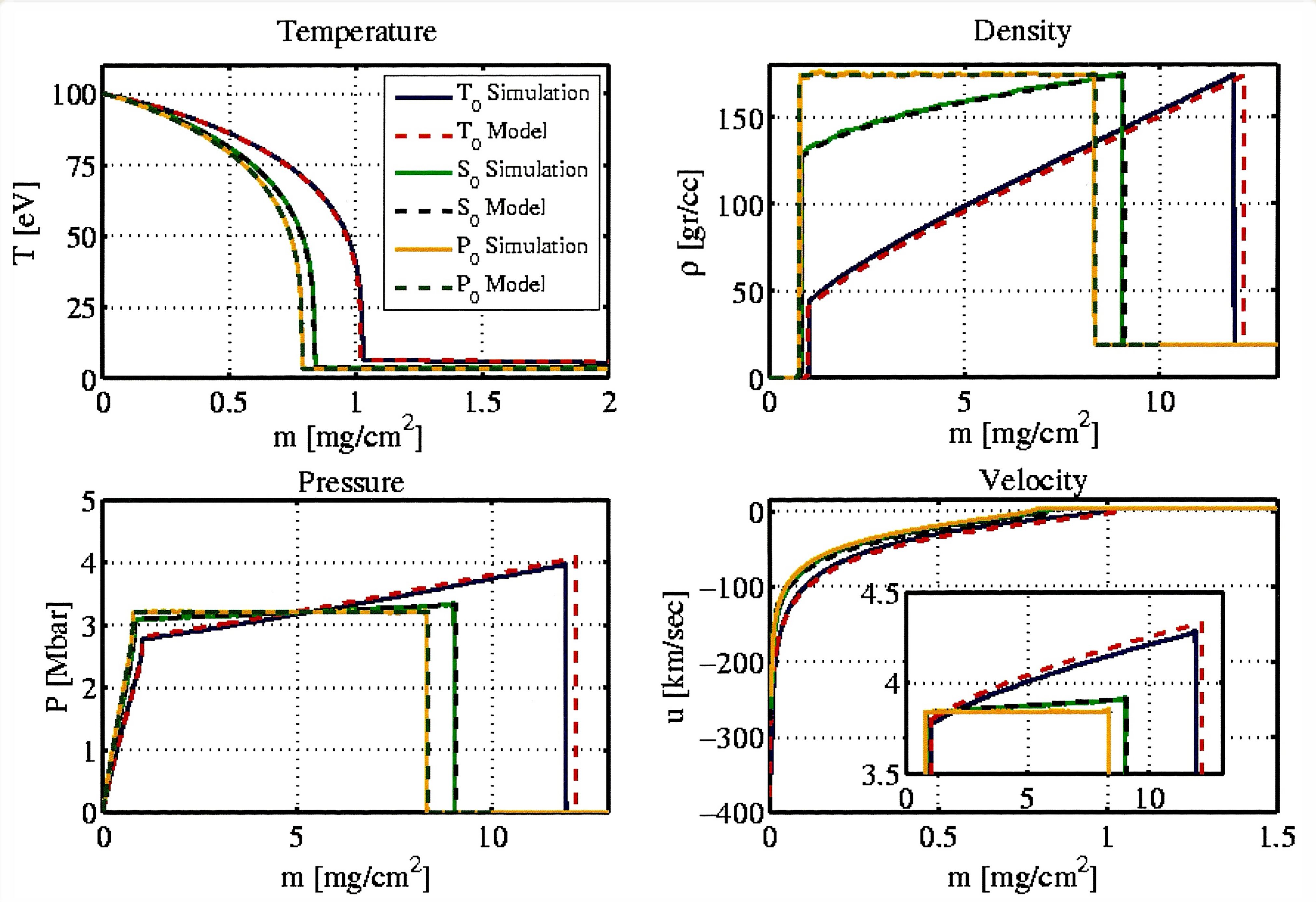
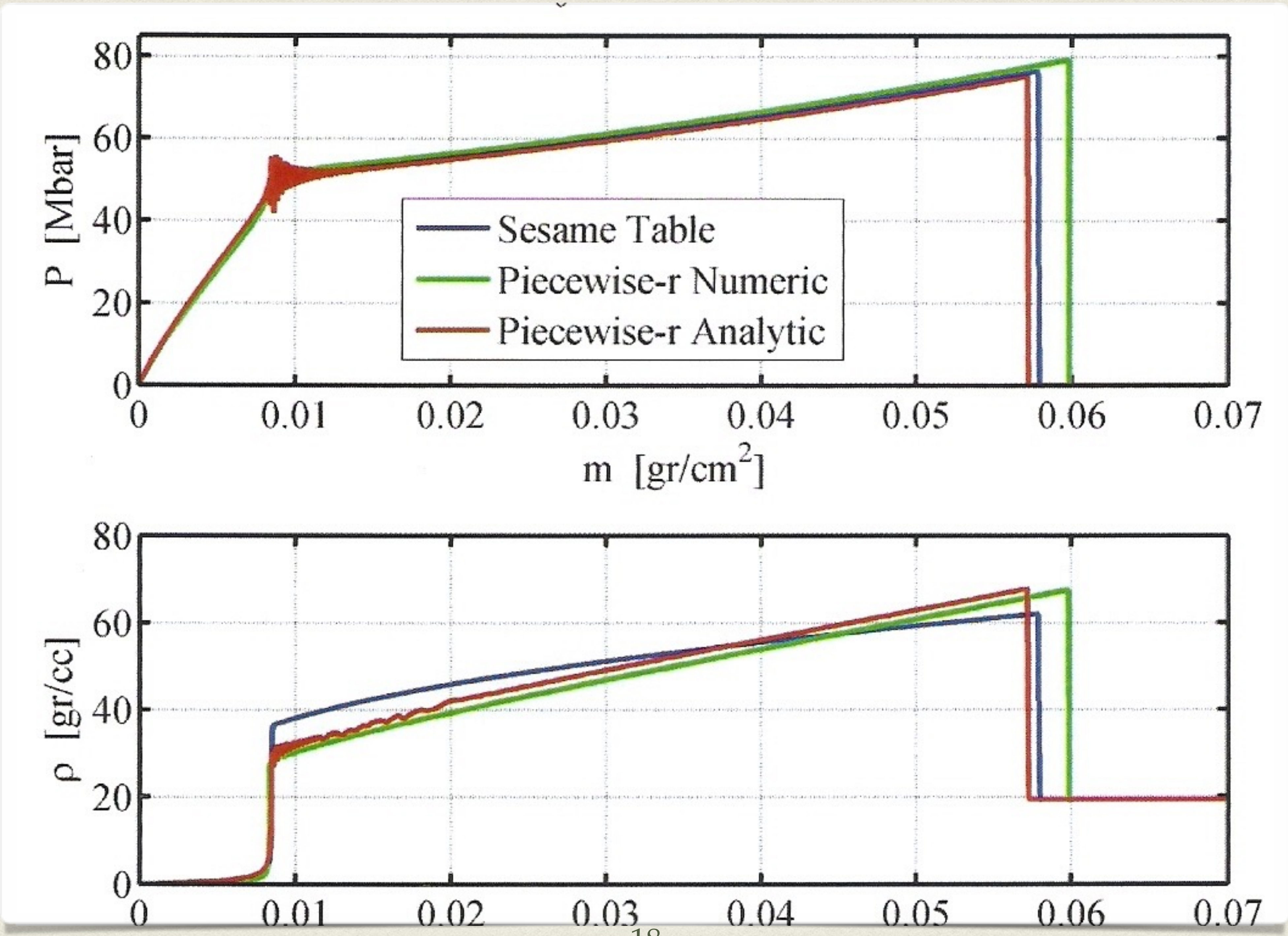


PIECEWISE EOS CALCULATION



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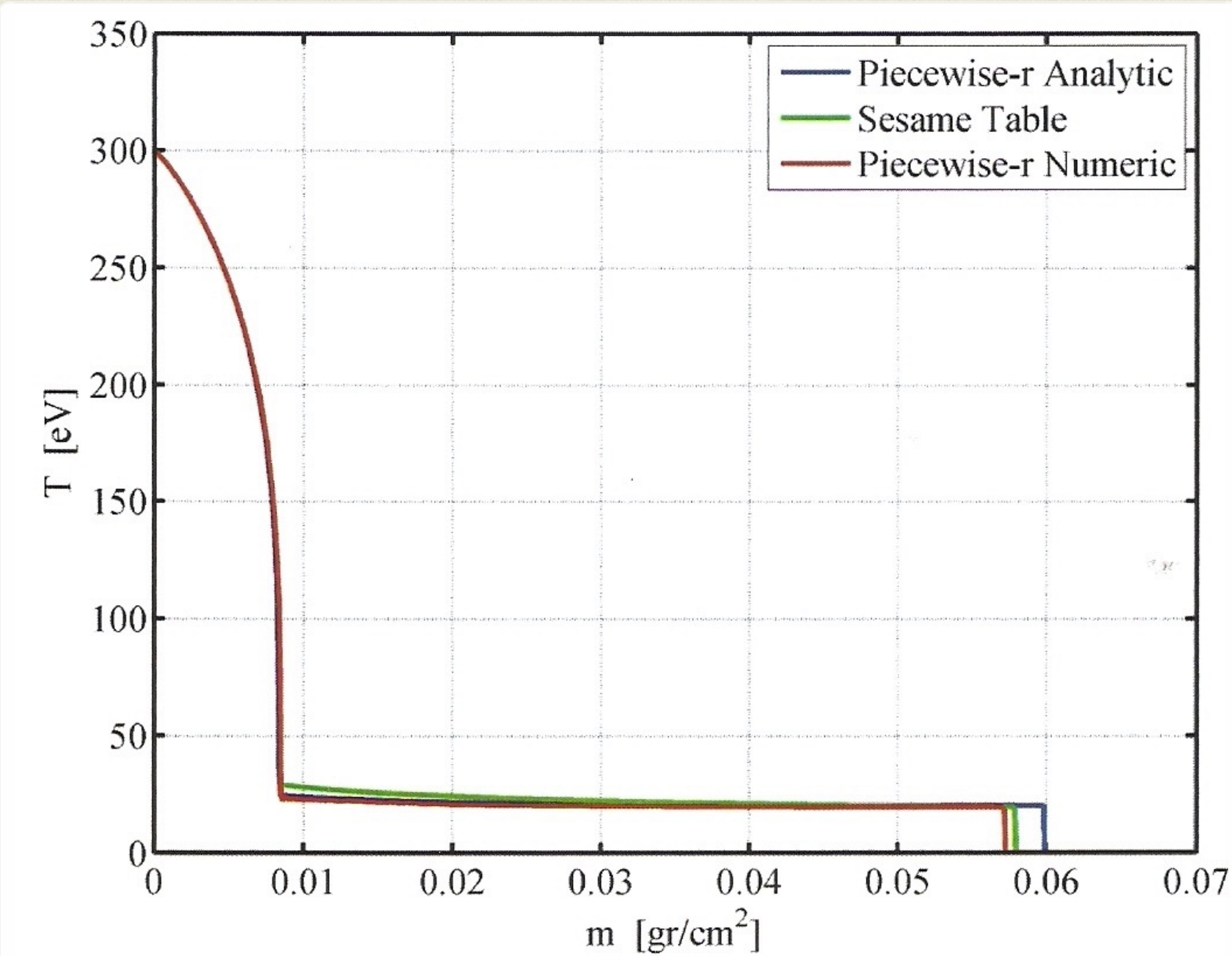

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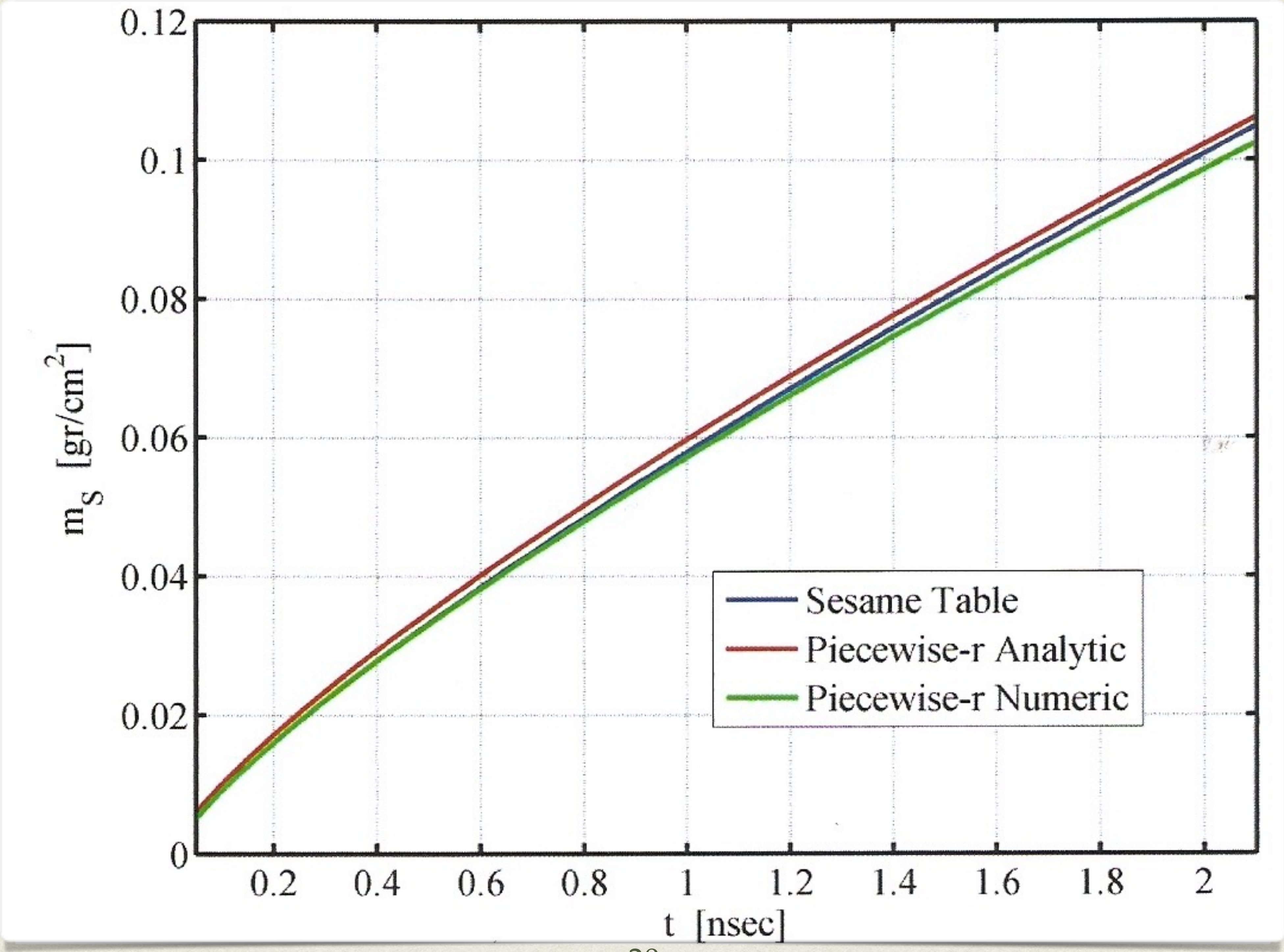

SUMMARY

- We provide a full solution to the subsonic heat wave for the general boundary power law condition.
- We solve the ablation region and shock region separately, and patch them using the right boundary conditions.
- The solutions allow for better planning of laser-plasma interaction experiments, and better understanding of ablation-driven shock waves.
- Details in Phys. Plasmas **22**, 082109 (2015)